

BEE 208 — Question 15: Repeated Ohm's Law Calculator

Control Statement: do...while loop | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables voltage, current, resistance (double) and continueChoice (char)

Step 3: Do:

Step 4: Prompt user to enter voltage and current

Step 5: Read voltage and current

Step 6: $\text{resistance} = \text{voltage} / \text{current}$

Step 7: Display resistance in ohms

Step 8: Ask user whether to continue (Y/N)

Step 9: Read continueChoice

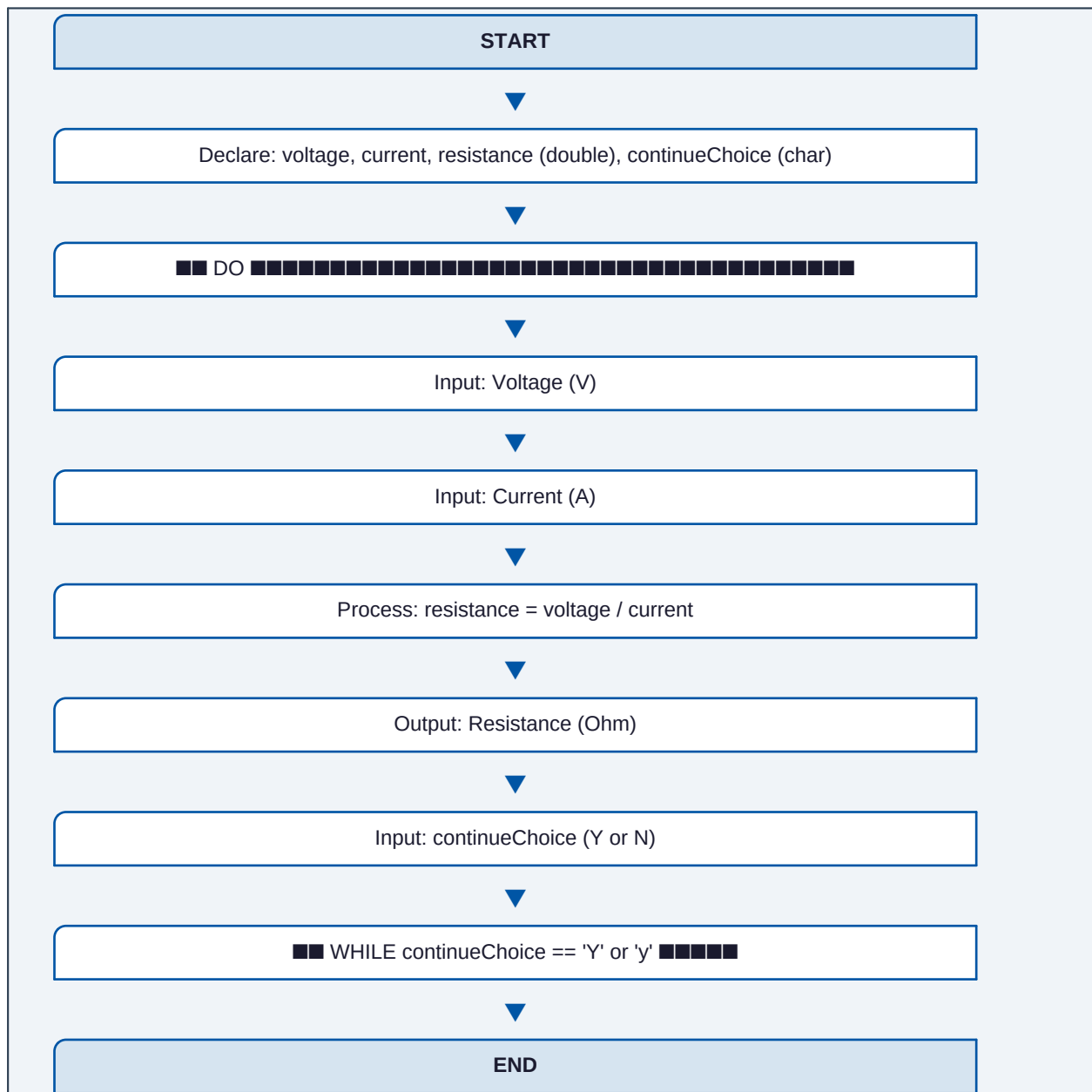
Step 10: While continueChoice == 'Y' or 'y'

Step 11: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE voltage, current, resistance AS DOUBLE  
  
DECLARE continueChoice AS CHAR  
  
DO  
  
  OUTPUT 'Enter Voltage (V):'  
  
  INPUT voltage  
  
  OUTPUT 'Enter Current (A):'  
  
  INPUT current  
  
  resistance = voltage / current  
  
  OUTPUT 'Resistance:', resistance, 'Ohm'  
  
  OUTPUT 'Do you want to continue? (Y/N):'  
  
  INPUT continueChoice  
  
  WHILE continueChoice == 'Y' OR continueChoice == 'y'  
  
END
```

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	V=12V, I=4A, continue=Y; V=24V, I=6A, continue=N	Expected Output	3 Ohm, then 4 Ohm
Input	V=230V, I=10A, continue=N	Expected Output	23 Ohm